

# 4

## Trade Size and Market Depth

Learning Objectives:

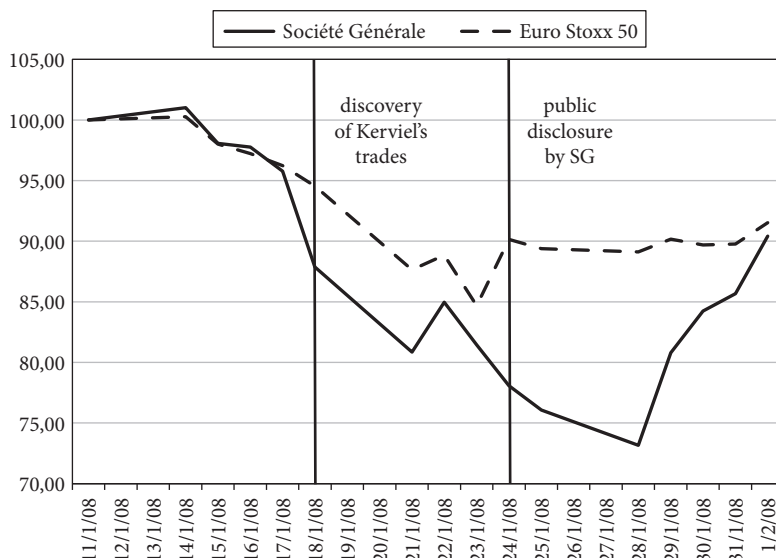
- How trade size affects prices
- What market depth is
- How traders choose their trade size
- Why the number of dealers is a determinant of market liquidity

### 4.1 Introduction

On Friday, January 18, 2008, the top management of Société Générale (the world's leading equity derivative trading house at the time) discovered that one of their traders, Jérôme Kerviel, had accumulated massive unauthorized long positions in European stock index futures (estimated at somewhere between forty and fifty billion euros), which had already accrued losses of about €1.5 billion. On Monday, January 21, the bank began liquidating these positions in a series of small trades and waited until January 24 to go public with the news, giving itself time to unwind the positions and avoid greater losses. Even so, by Friday, January 25, its losses amounted to €4.9 billion, and its stock price slid by 13% relative to the previous Friday's close, while the EuroStoxx price index (which comprises the 50 largest European stocks) dropped by 5%, as shown in Figure 4.1.<sup>1</sup> Yet Société Générale Chairman Daniel Bouton commented: "Had we not acted swiftly, the loss could have been ten times worse." As Société Générale was unwinding its positions, stock markets fell so sharply that Société Générale fire sales probably contributed to the stock market drop, even though the selling was limited to about 10% of daily trading volume.

This episode illustrates three facts. First, the use of a gradual sales strategy demonstrates that when an investor wants to sell a large amount of securities

<sup>1</sup> In the figure, the stock price of Société Générale and the EuroStoxx index are both standardized to 100 on January 11, 2008. Over the same week the S&P 500 was relatively stable, only falling by 0.4%. So the stock market drop was specific to the European stock market, in which Kerviel had taken gigantic exposures.



**Figure 4.1** Evolution of Société Générale stock price and EuroStoxx 50 Index from January 11, 2008, to February 1, 2008.

he should try to avoid very large orders that are likely to unsettle market prices. Second, the fact that Société Générale kept its exposure secret for a whole week while unwinding it suggests that the news would have had a serious adverse impact on market prices. Finally, the fact that the market turned down illustrates that even this gradual, covert strategy was not entirely successful at insulating the company from loss. In short, even very liquid markets—such as those for equity index futures—have limited depth, so large orders can unsettle prices.

In this chapter we present a framework for analyzing the determinants of market depth (i.e., how and why transaction prices react to order size). Recall that a deep market is one in which large orders do not have a much greater impact on prices than small orders. We shall see that depth depends on three of the factors that affect the bid-spread (see Chapter 3), namely (i) asymmetric information, (ii) risk aversion, and (iii) rents due to imperfect competition among market makers:<sup>2</sup>

<sup>2</sup> Order-processing costs do not affect market depth, as long as they are proportional to the number of shares traded. However, if they are more than proportional to trade size, they contribute to widen the bid-ask spread as trade size increases; conversely, if they are less than proportional, they tend to reduce the bid-ask spread as trades grow larger.

- (i) Insofar as larger orders reflect more private information than small orders, they induce larger price movements. The sensitivity of prices to trade size is then determined by the degree of asymmetric information between liquidity providers and informed traders: the greater the asymmetry, the shallower the market—that is, the more sensitive prices are to an order of a given size.
- (ii) Another factor affecting market depth is the risk aversion of liquidity providers. Filling a large order exposes providers to greater inventory risk; consequently the more risk-averse they are, the larger the price concession that they require to execute trades. To the extent that the order is absorbed by multiple liquidity providers rather than a single one, market depth will depend on their aggregate risk-bearing capacity, hence also on their number.
- (iii) Finally, strategic market makers have market power. As liquidity providers' market power is inversely related to their number, this is another reason why market depth increases with the number of liquidity providers.

We present these ideas in Section 4.2, building upon one of the most popular models of price formation under asymmetric information: the static version of the Kyle (1985) model of informed trading and market liquidity. The price impact of orders is increasing in order size, to an extent that depends on the informational advantage of the informed investor and the volume of liquidity or “noise” trading (trades that are not information driven). Section 4.2.4 extends the model to imperfectly competitive market-making, in the context of a call market (also known as a batch auction), where all traders submit supply or demand schedules and a Walrasian auctioneer crosses them at a common market-clearing price (see Chapter 1). Then, in Section 4.3, we analyze a call market where market makers have inventory concerns, being risk averse. We show that also in this setting the price impact of orders is increasing in order size, and depends on the underlying volatility of the stock and the risk aversion and market power of the market makers.

## 4.2 Market Depth under Asymmetric Information

To understand the determinants of market depth, we start with a model in which some traders have superior information. The logic is similar to that of the Glosten-Milgrom model described in Chapter 3. Here too, the price is set by risk-neutral market makers, but with two differences. First, orders can be of any size. Second, market makers may vary their prices depending on the quantities

traded. This extension enables the model to portray the role of trade size in conveying information and driving price movements.

The vehicle for our analysis is the model developed by Kyle (1985), now a standard for studying trading in the presence of asymmetric information. As in Glosten-Milgrom, the demand for liquidity comes from two types of traders, informed and uninformed. But whereas in that model, market makers faced either an informed or an uninformed order, here orders from the two types of traders are batched together. Uninformed investors submit a random aggregate order  $u$ , which is normally distributed, with zero mean and variance  $\sigma_u^2$ . The informed trader has advance knowledge of the security's value  $v = \mu + \varepsilon$ , which other market participants see as a variable drawn from a normal distribution with mean  $\mu$  and variance  $\sigma_v^2$ , following Kyle's notation (note that  $\sigma_v^2$  is simply the variance denoted by  $\sigma_\varepsilon^2$  in Chapter 3). Variables  $u$  and  $v$  are independent, and the informed trader does not know the uninformed order  $u$ . He can choose the size of his order and make it contingent on  $v$  by placing an order of size  $x = X(v)$ . Only market orders are allowed, not limit orders. The net batched order (the "order flow")  $q = x + u$  is submitted to the market. The market makers do not observe  $v$ , but infer it—imperfectly—from the order flow,  $q$ , or from the clearing price (depending on the way trading is organized).

In all the models considered below, the relationship between the equilibrium price and the order flow is linear and given by:

$$p = \mu + \lambda q. \quad (4.1)$$

From the standpoint of a trader who wants to buy or sell, the slope  $\lambda$  in this expression measures the price pressure exerted per unit of order size. More formally, this slope measures the price impact  $dp/dq$ . A market is deep when even a large order does not shift the price by much, that is, when  $\lambda$  is small. Formally, the depth of the market can be measured by  $1/\lambda$ , the inverse of the price pressure parameter, in other words, the amount of order flow that drives the price up by one unit.

The value of  $\lambda$  in equilibrium depends on how informative the order flow is (as it drives dealers' inferences about the payoff): in Section 4.2.1, we explain how market makers update their estimate of the security's value on observing the order flow. Section 4.2.2 considers a setting in which competition eliminates all market makers' rents, so that their price is equal to their estimate of the security's value on observing the order flow. In Section 4.2.3, we close the model by deriving both the optimal strategy of a monopolistic informed trader and the equilibrium order flow. This gives us a full description of market participants' strategies in equilibrium, as derived by Kyle (1985). Finally, in Section 4.2.4 we allow for market power in a call auction framework, where the strategic

interaction between market makers affects price determination, and market makers earn oligopoly rents.

#### 4.2.1 Learning from Order Size

Market makers know that the order flow for a stock reflects information about the fundamental value  $v$ , because one of the orders comes from the informed investor, who knows  $v$  in advance. But there are two obstacles to extracting even part of this information from the order flow: (i) the informed order  $x$  is overlaid by the noise of the order  $u$  of uninformed investors and (ii) market makers must guess how the informed investor's trade is related to his information. This guess must be right; if not, it would lead to systematic losses for market makers, so eventually we shall have to check whether informed investors really do follow the type of strategy that market makers expect. In other words, we have to solve for a rational expectations equilibrium, where market participants' conjectures are verified in equilibrium.<sup>3</sup>

Market makers will obviously expect the informed trader to buy when he knows that the true value  $v$  of the security is higher than other market participants' estimate  $\mu$ , and sell if it is lower. Specifically, suppose that they conjecture the informed investor to trade an amount  $x$  that is proportional to the discrepancy  $v - \mu$ :

$$x = X(v) = \beta(v - \mu) \quad (4.2)$$

for some parameter  $\beta > 0$ .<sup>4</sup> What the market makers actually observe, however, is not the informed trader's order alone, but the total net order flow, including the noise traders' orders:

$$q = x + u. \quad (4.3)$$

That is, the net order flow is a noisy signal of the asset value  $v$ , so market makers can use the noisy signal  $q$  to form their expectation of  $v$ . The resulting

<sup>3</sup> The concept of rational expectations equilibrium is an important tool to model situations of trading with asymmetric information; its application to competitive securities markets was developed by Grossman (1976), Hellwig (1980), Diamond and Verrecchia (1981), and Admati (1985), among others.

<sup>4</sup> In Section 4.2.3, we show that this conjecture is verified: there is a rational expectations equilibrium in which the optimal strategy of the informed investor is linear in  $v - \mu$ .

conditional expectation  $E(v|q)$  will generally differ from their unconditional expectation  $\mu$ .

As the aggregate order size from noise traders  $u$  is normally distributed and independent of  $v$ , the expected value of  $v$  conditional on  $q$  is provided by the ordinary least squares regression rule:<sup>5</sup>

$$\begin{aligned} E(v|q) &= \mu + \frac{\text{cov}(v, q)}{\text{var}(q)} \cdot q \\ &= \mu + \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2} \cdot q \\ &= \mu + \alpha q, \end{aligned} \quad (4.4)$$

where  $\alpha = \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2}$  measures the sensitivity of expectations to the order flow. That is,  $\alpha$  measures how informative the order flow is.

The informativeness of the order flow,  $\alpha$ , depends on  $\sigma_u^2$ ,  $\sigma_v^2$ , and  $\beta$ . For a fixed value of  $\beta$ , a decrease in the variance of noise trading  $\sigma_u^2$  makes the order flow a more precise signal of the asset's value, and thereby enhances its informativeness. By the same token, informativeness increases with the variance of the security's value  $\sigma_v^2$ , since this increases the fraction of order flow volatility that can be attributed to the informed trader, and makes the total order flow more informative. The impact of  $\beta$ —the aggressiveness of the informed trader—is non-monotonic. Specifically, it is increasing for small values of  $\beta$  and then decreasing, because an increase in  $\beta$  has two opposite effects: greater aggressiveness by informed traders tends to make the order flow more informative, but at the same time inflates trades for any given value of  $v$ , so that the dealer must correspondingly scale down the extent to which he adjusts his estimate of  $v$  upon observing any given order flow. The first effect dominates for small values of  $\beta$ ; the second dominates for sufficiently large  $\beta$ .

<sup>5</sup> Consider two joint normally distributed random variables, and:

$$X \sim N(\mu_x, \sigma_x^2) \text{ and } Y \sim N(\mu_y, \sigma_y^2),$$

and denote their covariance  $\sigma_{xy}$ . A property of the bivariate normal distribution is that the conditional density of  $Y$  given  $X = x$  is itself normal with conditional mean:

$$E(Y|x) = \mu_y + \frac{\sigma_{xy}}{\sigma_x^2} (x - \mu_x) = \left( \mu_y - \frac{\sigma_{xy}}{\sigma_x^2} \mu_x \right) + \frac{\sigma_{xy}}{\sigma_x^2} x,$$

which is the predicted value of  $Y$  from an ordinary least squares (OLS) regression of the equation  $Y = a + bX$ , upon setting the explanatory variable  $X = x$ . The slope coefficient  $\sigma_{xy}/\sigma_x^2$  is precisely the OLS estimate of  $b$ . In our case, the dependent variable is  $v$  and the explanatory variable is  $q$ .

### 4.2.2 Perfectly Competitive Dealers

We now consider the case in which dealers are risk-neutral and trading is organized in three steps. In the first step, market orders from the two types of trader are batched into a single net order  $q = x + u$ . Second, each market maker responds with the price at which he is willing to execute this order in full. Third, the entire batched order is routed to the market maker who posts the best price.<sup>6</sup> Effectively, this means that dealers engage in Bertrand competition.

The expected profit of the dealer who wins the order at price  $p$  is:

$$E[q(p - v)|q] = q[p - E(v|q)].$$

Competition between dealers drives this expected profit down to zero. For instance, suppose that  $q > 0$  and that  $p > E(v|q)$ . If a dealer just matches the best offer,  $p$ , he is not sure he will execute the order, which will be allocated among all the bidders at price  $p$ . So he is strictly better off by slightly undercutting his competitors to win the order. This continues until the best offer is driven down to the dealers' conditional estimate of the security's value (4.4). At this point, it is no longer profitable for anyone to undercut the best offer and an equilibrium is reached. Hence, in this case, the equilibrium price is:

$$p = E(v|q) = \mu + \alpha q, \quad (4.5)$$

and therefore:

$$\lambda = \alpha = \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2}. \quad (4.6)$$

Thus, market depth is entirely determined by the informativeness of the order flow. The less informative the order flow, the deeper the market. The intuition is straightforward. In this model, dealers use the order flow as a signal of the asset's value and adjust their prices accordingly. If the order flow is not very informative, it does not greatly affect dealers' value estimate, which means that it has little impact on prices since quotes are competitive.

Consider now a trader placing an order of size  $u$ : what price can he expect to get? He obviously knows his own order size  $u$ ; he knows nothing about the size of other orders, but expects that, on average, they net out to zero. Therefore, based on equation (4.5), he expects his own order to be executed at price

<sup>6</sup> If several post the same price, the order is allocated to one of them.

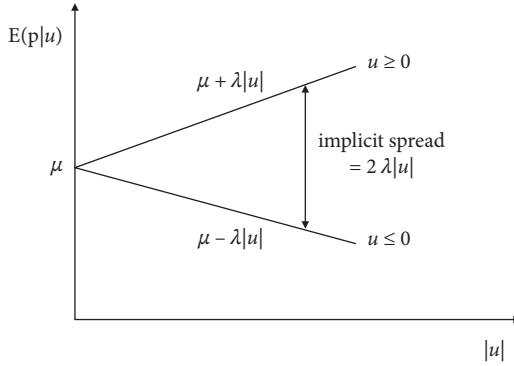


Figure 4.2 Depth and effective bid-ask spread.

$E(p) = \mu + \lambda E(q|u) = \mu + \lambda u$ . Figure 4.2 plots the relationship between this price and the investor's average order size,  $|u|$ .

Relative to traders' prior estimate of the asset payoff ( $\mu$ ), buy market orders execute at a premium and sell market orders at a discount equal to  $\lambda|u|$ . Thus,  $\lambda|u|$  is the effective half bid-ask spread paid by investors on a transaction of size  $|u|$ . As Chapter 2 explains, the notion of effective bid-ask spread and market depth are closely related: the effective spread increases with the size of the order at a rate of  $\lambda$ .

What does this model tell us about the volatility of stock returns? We can think of the total return as  $v - \mu$ , the difference between the security's final value  $v$  and investors' pre-trading valuation  $\mu$  (its initial market price). This total return can be split into two components: an initial return up to the time of trading,  $p - \mu$ , and a subsequent return from the time of trading to the moment when the final value  $v$  is revealed,  $v - p$ .

The volatility of the initial return is:

$$\text{var}(p - \mu) = \lambda^2(\beta^2 \sigma_v^2 + \sigma_u^2) = \frac{\beta^2 \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2} \sigma_v^2, \quad (4.7)$$

which is increasing in the aggressiveness of informed trading,  $\beta$ . An increase in  $\beta$  means larger trade sizes for the informed trader (the "trade size effect"); thus, for any fixed value of  $\lambda$ , prices deviate more from traders' prior value estimate. This effect is partially offset by the fact that  $\lambda$  decreases with  $\beta$ , for large enough  $\beta$ . But this latter effect is always outweighed by the trade size effect. In short, trading is a source of volatility, because the order flow contains information that moves the price.

The volatility of the return after trading,  $v - p$ , is:

$$\text{var}(v - p) = \frac{\sigma_u^2}{\beta^2 \sigma_v^2 + \sigma_u^2} \sigma_v^2, \quad (4.8)$$

which is decreasing in the informed trader's aggressiveness,  $\beta$ . Intuitively, this is because if the informed investor trades more aggressively, more of his information is reflected in the market price, so the price is closer to the fundamental  $v$ . Price discovery is faster. In this model  $\text{var}(v - p)$  equals the "average pricing error," which Chapter 3 defines as  $E[(v - p)^2]$ , and presents as an inverse measure of price discovery.<sup>7</sup> This finding parallels our result in Chapter 3 that an increase in the fraction of informed traders accelerates price discovery.

The variance of the total return on the security over both periods (before and after trading) is:

$$\text{var}(v - \mu) = \text{var}(v - p) + \text{var}(p - \mu) = \sigma_v^2, \quad (4.9)$$

since  $\text{cov}(v - p, p - \mu) = 0$ .<sup>8</sup> It is important to note that even though informed trading affects the variance of the pre- and post-trade returns ( $p - \mu$  and  $v - p$ ), it leaves the volatility of the total return ( $v - \mu$ ) unchanged. Informed trading does not change the fundamental risk of the security: it simply brings forward the resolution of part of the uncertainty from the final date when  $v$  is revealed to the date when trading occurs.

### 4.2.3 The Informed Trader's Order Placement Strategy

So far, we have taken the trading strategy of the informed trader as given. Now we endogenize his behavior. As in Kyle (1985), we posit that dealers are competitive (so that they obtain no rents), and that the informed trader is risk neutral.

Intuitively, the informed trader's strategy depends on the dealers' pricing policy: the informed investor should trade less aggressively on his information when he expects dealers' prices to be very sensitive to the order flow. In turn, as we have seen, dealers' strategies themselves depend on the informed trader's aggressiveness. Thus, the strategies of dealers and of the informed trader are interdependent. In a Nash equilibrium, each agent behaves optimally given the other agents' behavior. We show that the following strategies constitute a Nash equilibrium:

<sup>7</sup> Here the average pricing error is equal to  $\text{var}(v - p)$ , since  $E(v - p) = 0$ .

<sup>8</sup> This covariance is zero because the price is the best estimate of the fundamental  $v$  at the time of trading, so that  $v - p$  is the unanticipated component of the fundamental (the portion not yet reflected in the price), and thus uncorrelated with the price  $p$  itself.

$$\text{dealers' quote: } p(q) = \mu + \lambda q, \text{ with } \lambda = \frac{\sigma_v}{2\sigma_u}, \quad (4.10)$$

$$\text{informed investor's trading: } X(v) = \beta(v - \mu), \text{ with } \beta = \frac{\sigma_u}{\sigma_v}. \quad (4.11)$$

**Informed investor.** Given dealers' pricing strategy (equation (4.10)), the informed investor's expected profit if he trades  $x$  shares is:

$$E[(v - p)x] = E[(v - \{\mu + \lambda(x + u)\})x] = (v - \mu - \lambda x)x.$$

Thus, the order size that maximizes his expected profit is:

$$X(v) = \beta(v - \mu), \quad (4.12)$$

with:

$$\beta = \frac{1}{2\lambda}. \quad (4.13)$$

The trade size of the informed investor ( $|X(v)|$ ) is inversely related to  $\lambda$ . Intuitively, the informed investor will place larger orders in deeper markets because they have less adverse impact on prices.

**Dealers.** From Section 4.2.1, we know that if dealers expect the informed investor to follow a linear strategy  $X(v) = \beta(v - \mu)$  (equation (4.2)), then their competitive price, for each value of  $q$ , is given by:

$$p(q) = \mu + \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2} q. \quad (4.14)$$

Thus, we have:

$$\lambda = \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2} \Rightarrow \frac{1}{\lambda} = \beta + \frac{\sigma_u^2}{\beta \sigma_v^2}. \quad (4.15)$$

Combining (4.13) and (4.15) yields the equilibrium values of  $\lambda$  and  $\beta$ :

$$\lambda = \frac{\sigma_v}{2\sigma_u}, \beta = \frac{\sigma_u}{\sigma_v}.$$

The larger the variance of noise trading  $\sigma_u^2$  or the lower the variance of the fundamental value  $\sigma_v^2$ , the deeper is the market in equilibrium. Intuitively, the greater  $\sigma_u^2$ , the greater the average order flow in the market, and the less likely that an order of any fixed size will be interpreted as information driven. The smaller

$\sigma_v^2$ , the smaller the informed investor's average informational advantage, and the less dealers need to protect themselves against it.

In equilibrium, dealers absorb the aggregate market order. Thus, their net supply is  $q$ , and their expected profits can be written as:

$$E[(p - v)q | q = x + u] = E[(p - v)x | q = x + u] + E[(p - v)u | q = x + u].$$

The first term in this decomposition is the expected profit on trades with the informed trader; the second is the expected profit on trades with uninformed investors. As the informed investor has an informational advantage, dealers lose on these trades, so that  $E[(p - v)x | q = x + u] < 0$ . As in equilibrium, dealers just break even, they must recover this loss by the expected profits on trades with the uninformed investors:

$$E[(p - v)x | q = x + u] = -E[(p - v)u | q = x + u].$$

Since this equality must hold for all possible realizations of  $q$ , it must also hold unconditionally (by the law of iterated expectations):

$$E[(v - p)x] = E[-(v - p)u].$$

Hence, the expected profit of the informed investor is just equal to the expected losses of the uninformed, who ultimately bear the adverse selection costs (as explained in Chapter 3).

In equilibrium, the informed investor's expected profit, conditional on a realized value of  $v$ , is:

$$\begin{aligned} E[(v - p) \cdot X(v) | v] &= [v - \mu - \lambda E(q | x)] \cdot x \\ &= [v - \mu - \lambda x] \cdot x = \frac{1}{2} \frac{\sigma_u}{\sigma_v} (v - \mu)^2, \end{aligned}$$

since  $x = \frac{\sigma_u}{\sigma_v} (v - \mu)$ . Therefore, his unconditional expected profit is:

$$E[(v - p) \cdot x] = \frac{1}{2} \sigma_u \sigma_v.$$

Thus the informed investor has greater expected profit when there is more uninformed trading ( $\sigma_u$ ) or when he has a larger informational advantage ( $\sigma_v$ ). A greater volume of uninformed trading, in fact, provides more camouflage, since the aggregate order flow is a noisier signal than the informed investor's actual trade.

Recall that in this model, we can measure price discovery by the average pricing error,  $\text{var}(v - p) = E[(v - p)^2]$  (see Section 4.2.2). Given the equilibrium actions of the informed investors and the dealers, price discovery is:<sup>9</sup>

$$E[(v - p)^2] = \frac{1}{2}\sigma_v^2. \quad (4.16)$$

Thus, in equilibrium, half the uncertainty about the security's value is resolved at the time of trading.

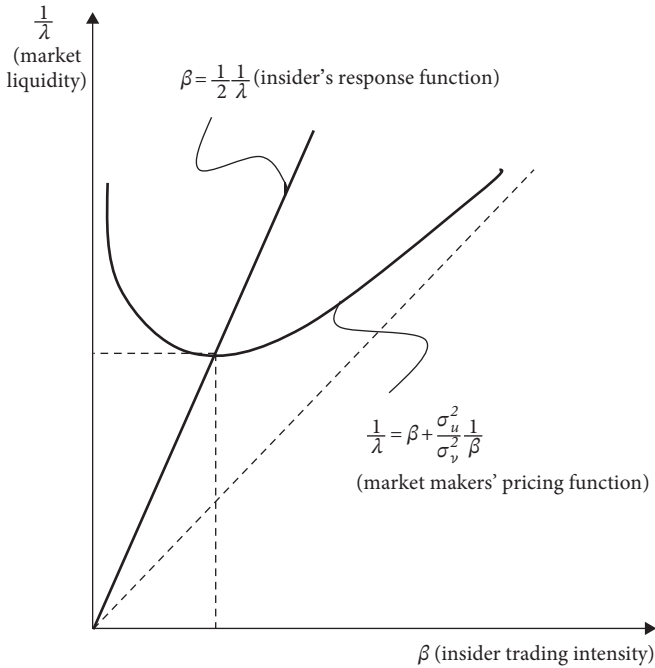
An instructive way to represent the model graphically is to draw the two relationships that relate  $\beta$  and  $\lambda$ : equation (4.13), which gives the informed investor's choice of trading intensity  $\beta$  as an increasing function of the depth of the market  $1/\lambda$ ; and equation (4.15), which shows the market maker's choice of liquidity  $1/\lambda$  as a convex function of the informed investor's trading intensity. This second relationship is non-monotonic because the market maker can afford to provide a deep market both when the informed investor's trading activity is very low *and* when the informed investor trades so aggressively as to give away his presence on the market. These two functions are plotted in Figure 4.3. The intersection indicates the Nash equilibrium point, which happens to coincide with the point where market depth is minimized ( $1/\lambda$  is at its lowest point when  $\beta$  equals  $\sigma_u/\sigma_v$ ).

Depth improves whenever the informed investor's trading intensity differs from the level portrayed in Figure 4.3: the line representing the informed investor's response function will then swivel away from the position shown in the figure. This occurs, for instance, if the informed investor is exposed to competition from other informed investors or is risk averse. (We leave the analysis of these two cases to the reader in exercises 3 and 6.) This means that, unlike the simple situation shown in Figure 4.3, the equilibrium may actually be to the left or to the right of the minimum-liquidity point. This suggests to an important caveat for anti-insider-trading regulation and enforcement. In practice, one may not know whether the market equilibrium is to the left or to the right of the minimum-depth equilibrium point, so it is hard to say whether cracking down on insider trading, which steepens the insiders' response function in Figure 4.3, would increase or decrease market liquidity.

<sup>9</sup> We obtain equation (4.16) as follows. We have:

$$v - p = v - \mu - \frac{\sigma_v}{2\sigma_u}(x + u) = \frac{v - \mu}{2} - \frac{\sigma_v}{2\sigma_u}u,$$

since in equilibrium,  $x = (\sigma_u/\sigma_v)(v - \mu)$ . Computing the variance and observing that  $v$  and  $u$  are independent, we obtain the result in the text.



**Figure 4.3** Insider's behavior and pricing function in Kyle's model.

To conclude this section, let us discuss some possible extensions. First, we have assumed that the informed investor can trade only once, after which his information becomes publicly known, hence obsolete. This means that he does not worry that by trading aggressively today he alerts the market to his presence and so lowers his potential future gains. Kyle (1985) analyzes the trade-offs involved in a multiperiod version of the model. In keeping with intuition, he shows that the informed investor trades less aggressively than in the one-shot case, in order to avoid dissipating his information advantage too quickly. In this case, information is incorporated into market prices only gradually, as the result of the repeated trading by the informed investor. In the limiting case of trading in continuous time, the rate at which information gets embedded in market prices is constant, and so is the volatility of price changes per unit of time.

Second, we have assumed that there is only one informed investor. The static model can easily be extended to consider multiple informed investors (see exercise 3). In this case, the depth of the market increases with the number of informed investors. Holden and Subrahmanyam (1992) consider the multiperiod case and show that competition among informed investors results in a race that dissipates their informational advantage very quickly by comparison with the case of a single informed investor.

#### 4.2.4 Imperfectly Competitive Dealers

The trading mechanism considered in Sections 4.2.2 and 4.2.3 is such that competition among dealers results in zero expected profits for the dealers. This is because risk-neutral dealers submit price bids for the *total* quantity demanded, and the auctioneer selects the best bid to fill the entire net order flow  $q$ . We now consider a different trading mechanism, namely a call auction (as described in Chapter 1), in which each dealer submits a schedule of offers that specifies the number of shares he is willing to buy or sell at each possible price, without knowing other dealers' offers. An auctioneer then parcels out investors' aggregate demand,  $q$ , among the dealers according to their demand or supply at the price that clears the market. Specifically, let  $Y^k(p)$  be the total number of shares that dealer  $k$  is willing to supply at price  $p$ : if  $Y^k(p) > 0$ , the dealer is willing to sell shares; if  $Y^k(p) < 0$ , he is willing to buy  $|Y^k(p)|$  shares. If there are  $K$  dealers, their aggregate supply at price  $p$  is  $\sum_{k=1}^K Y^k(p)$ . Hence, when the aggregate investor demand for the security is  $q = x + u$ , the clearing price  $p^*$  set by the auctioneer is given by the market-clearing condition:

$$\sum_k Y^k(p^*) = q. \quad (4.17)$$

We shall see that with this trading mechanism, competition among dealers does not drive their expected profits to zero unless their number is very large. Moreover, in this setting market depth depends not only on the informativeness of the order flow (as in Section 4.2.2), but also on the number of dealers (which determines their market power). Hence, even fine details of trading arrangements can affect market makers' profits, and market liquidity.

Recall that in Section 4.2.2, the starting point was the market makers' conjecture about the trading strategy of the informed trader. Here, each market maker has to form a conjecture not only about the informed trader's strategy, but also about the behavior of the other market makers. This is because now, in contrast with the model of Section 4.2.2, he cannot take the price as given: he knows he can affect it by changing his own supply of the security, and the response of the equilibrium price depends on how his competitors react to his own action.

Thus, in order to determine the optimal behavior of any dealer, we must specify his beliefs about (i) informed investor trading ( $x$ ) and how that affects his estimate of the security's value based on total customer demand ( $q$ ), and (ii) the strategic behavior of his competitors, and what that means for the response of the equilibrium price to his own supply of the security. Suppose that because of informed trading, the best estimate of the security's value, given the total order flow  $q$ , is  $E(v|q) = \mu + \alpha q$ , as in Section 4.2.2. Suppose further

that each dealer  $k$  will supply shares according to the linear supply function  $Y^k(p) = \phi(p - \mu)$ . These two conjectures, together with market-clearing condition (4.17), enable each dealer  $k$  to identify a “residual demand function” (i.e., net customer demand  $q$  minus the supply from his  $K-1$  competitors for any price  $p$ ). He can then compute how the equilibrium price  $p^*(q)$  will respond to any possible supply of shares  $y^k$  he may decide on, and therefore identify his profit-maximizing supply function (his own best response)  $Y^k(p)$ .

Note that each dealer's best response  $Y^k(p)$  depends also on his estimate of the security's fundamental value  $E(v|\Omega^k)$  conditional on his  $\Omega^k$ . One would think that this information does not contain the aggregate market order  $q$ , since dealers do not directly observe  $q$  in the call auction and hence they cannot use it to estimate the fundamental value. However, they can infer  $q$  from the clearing price, since the clearing price depends on it, by condition (4.17). As dealers submit price-contingent orders, they can take into account the information contained in the clearing price when they bid. Thus, for each possible price  $p$ , their estimate of the value of the security is  $E(v|p^*(q) = p)$ , and they optimally choose the quantity  $Y^k(p)$  to buy or sell at this price given this estimate and other dealers' offers.<sup>10</sup> Then, noticing that by symmetry all dealers must choose the same supply function and imposing the equilibrium condition (4.17), one finds the value of the supply parameter  $\phi$  and the relationship  $p^*(q)$  consistent with the beliefs assumed. In other words, as in Section 4.2.2, here we also seek a rational expectations equilibrium.

In the context of this model, a rational expectations equilibrium is a set of schedules,  $\{Y^k(p)\}_{k=1}^K$ , for the dealers and a price mapping  $p^*(q)$  such that: (i) the schedule of offers posted by each dealer maximizes his expected profit, given his belief about the security's value and the price schedules chosen by his competitors; (ii) dealers correctly anticipate that the clearing price is given by  $p^*(q)$  and form their beliefs accordingly; and (iii) for each value of  $q$ , the market clears for  $p^*(q)$ . In this equilibrium, dealers' expectations are rational in the sense that their beliefs are based on the correct relationship between the clearing price and order flow  $q$  (the only informative variable about  $v$ ).

In Appendix A, we show that, if  $K \geq 3$ , there is a rational expectations equilibrium in which:

$$Y^k(p) = \frac{1}{\alpha} \frac{K-2}{K(K-1)} (p - \mu) \quad \forall k \in \{1, \dots, K\}, \quad (4.18)$$

<sup>10</sup> The idea that the clearing price conveys information and that traders should take this information into account in formulating their price-contingent orders is a key insight of rational expectations models such as Grossman (1976).

and the equilibrium price function is:

$$p^*(q) = \mu + \lambda q, \quad (4.19)$$

with  $\lambda = \alpha \frac{K-1}{K-2}$  and  $\alpha$  defined as in equation (4.4). The equilibrium is linear since both dealers' bidding strategies and the clearing price are linear functions.<sup>11</sup> Note that for equilibrium to exist, there must be at least three market makers. As expected, the market-clearing price reveals information on the payoff  $v$ , since it is a linear function of the aggregate order size. In equilibrium, the dealers' estimate of the value of the security when they trade at price  $p$  is  $E(v|p^*(q) = p) = E(v|q) = \mu + \alpha q$  (the first equality here reflects the fact that there is a one-to-one mapping between the price  $p$  and the aggregate market order size  $q$ ).

When  $K$  goes to infinity, we have  $\lambda = \alpha$ : if the number of dealers is very large, the depth of the market is entirely determined by the informativeness of the order flow, exactly as in Section 4.2.2. As dealers trade at the expected value conditional on their information, they obtain zero expected profits. We refer to this polar case as the competitive case.

In contrast, when the number of dealers is finite,  $\lambda$  is strictly larger than  $\alpha$ . Thus, dealers obtain strictly positive expected profits. In fact, we can write equation (4.19) as:

$$p = E(v|q) + \underbrace{\frac{1}{K-2}\alpha q}_{\text{markup}} = \mu + \alpha q + \frac{1}{K-2}\alpha q.$$

Under imperfect competition the dealers' price is higher than their valuation of the security, as illustrated in Figure 4.4 below: when the total order size is  $q$ , dealers earn an expected profit per share  $p - E(v|q) = \frac{1}{K-2}\alpha q > 0$ .

In other words, the trading rules in the call auction enable dealers to compete less aggressively than when they must post a single offer for the entire order, as was assumed in the previous section. The reason is as follows. The risk of trading against an informed investor leads dealers to limit the number of shares that they offer to buy or sell at any given price. As a result, each dealer has pricing power. To see this, suppose that in the aggregate investors want to be net buyers ( $q > 0$ ). For a given market order size, a dealer can raise the clearing price by reducing his supply, very much as a monopolist would do, since the other dealers supply only a limited number of shares at each price. But this possibility is limited by

<sup>11</sup> There might be other equilibria in which dealers' bidding strategies or the mapping  $p^*(q)$  are non-linear.

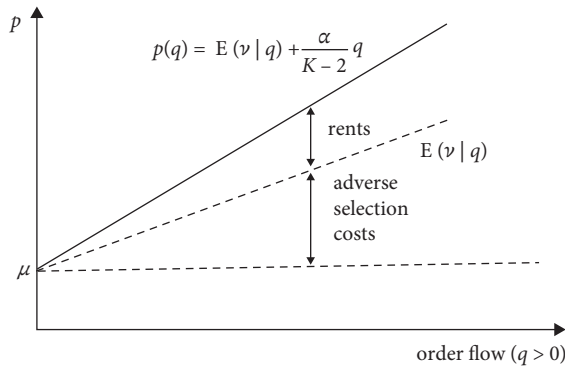


Figure 4.4 Market depth with imperfect competition.

the extent to which his competitors react by increasing their supply. Intuitively, if a small increase in price generates a large increase in other dealers' aggregate supply,<sup>12</sup> then a dealer must shave his own supply by a large amount in order to have a substantial price effect. In this case, the dealer benefits less from the increase in price by reducing his own sale volume. This happens when  $K$  is large and the equilibrium price is therefore close to the dealers' valuation  $E(v|q)$ . In contrast, when  $K$  is small, each dealer has significant market power because the others' aggregate supply response at each price is small. As a result, the wedge between the equilibrium price and dealers' valuation becomes large in markets with few dealers.<sup>13</sup>

For this reason, the depth of the market  $1/\lambda$  depends not only on the informational content of the order flow  $\alpha$  but also on the number of competing market makers  $K$ , as Figure 4.4. shows. More dealers tend to mean lower rents and a deeper market. Thus, liquidity providers' market power is a source of market illiquidity.

In this section, we have taken the informed investor's trade size as given. It is straightforward to endogenize this trade size as we did in Section 4.2.3 (see exercise 8). The behavior of the informed trader is unchanged, except that his trade size now increases with the number of dealers.

<sup>12</sup> That is, if the price elasticity of other dealers' aggregate supply is high.

<sup>13</sup> No linear equilibrium exists when  $K = 2$  because in this case, by reducing his supply by a small amount, the dealer obtains an infinitely large increase in price. When  $K = 1$ , the dealer can charge an infinite price to sell  $q$  shares, since the investors' aggregate demand is price insensitive.

### 4.3 Market Depth with Inventory Risk

As we saw in Chapter 3, the need to compensate market makers for holding risky positions is another determinant, in addition to adverse selection, of the bid-ask spread. Here we extend the analysis of price formation with inventory risk to a situation where investors interact via a call auction mechanism where they can place orders of any size. For simplicity, we suppose that there is no informed trading ( $x = 0$  so that  $q = u$ ). First, in Section 4.3.1, we posit that risk averse dealers are price-takers (that is, they behave competitively) in the call auction market. In this case, we obtain the same relationship between the equilibrium price and dealers' aggregate inventory as in Section 3.5.1. Then, in Section 4.3.2, we consider risk averse dealers who act strategically in the auction; that is, they realize that the clearing price depends on the way in which they themselves bid. As in the model with asymmetric information, we find that market depth is lower in this case than when dealers behave competitively.

#### 4.3.1 Perfectly Competitive Dealers

Consider again the inventory model of Section 3.5. The main differences between that model and the model analyzed in this section are that now orders may be of any size and trading is assumed to occur via a call auction, as in Section 4.2.4. In the auction, each dealer submits a price schedule  $Y^k(p)$  that specifies the number of shares he is willing to buy ( $Y^k(p) < 0$ ) or sell ( $Y^k(p) > 0$ ) if the clearing price is  $p$ . For simplicity, we further assume that there is a single trading round (at time  $t = 0$ ) and the asset payoff,  $v$ , is realized at its termination (time  $t = 1$ ). In this section, we assume that dealers behave competitively. That is, they neglect the impact of their orders on the equilibrium price (they are "price-takers").

There are  $K$  dealers, each with mean-variance preferences. Dealer  $k$  has risk-aversion coefficient  $\rho^k > 0$  and his expected utility is:

$$U^k = E(w_1^k) - \frac{\rho^k}{2} \text{var}(w_1^k),$$

where  $w_1^k$  is the mark-to-market value of the dealer's portfolio at date 1, that is,

$$w_1^k = v \cdot z_1^k + c_1^k = v(z_0^k - y^k) + c_0^k + p y^k,$$

where  $z_t^k$  is dealer  $k$ 's inventory in the risky security and  $c_t^k$  is his cash position at time  $t$ . If at time  $t = 0$  the dealer sells  $y^k$  shares, this decreases his risky inventory

by  $y^k$ , and increases his cash position by  $py^k$ . Hence,  $z_1^k = z_0^k - y^k$  and  $c_1^k = c_0^k + py^k$ , as shown in the previous equation. As in Chapter 3, the first-order condition for the maximization of dealer  $k$ 's expected utility is  $dU^k/dy^k = 0$ , which in the current setting implies:<sup>14</sup>

$$p - \mu + \rho^k \sigma_v^2 (z_0^k - y^k) = 0. \quad (4.20)$$

Thus, the optimal price schedule for dealer  $k$  is:

$$Y^k(p) = \frac{p - \mu}{\rho^k \sigma_v^2} + z_0^k. \quad (4.21)$$

Each market maker  $k$ 's price schedule is related inversely to his risk aversion  $\rho^k$  and directly to his initial inventory  $z_0^k$ .

As in Section 4.2.4, market-clearing requires that aggregate net sales by dealers equal the net market orders placed by customers:

$$\sum_{k=1}^K Y^k(p) = q. \quad (4.22)$$

In this setting, market orders will typically be filled by several dealers, each taking a portion, with some or all customer orders being directly crossed.

Inserting the supply functions (4.21) into the market clearing condition (4.22) yields:

$$\sum_{k=1}^K \left( \frac{p - \mu}{\rho^k \sigma_v^2} + z_0^k \right) = q,$$

so that:

$$p = \mu - \rho \sigma_v^2 Z + \rho \sigma_v^2 q, \quad (4.23)$$

where  $Z$  denotes the aggregate initial inventory of the market makers  $Z = \sum_{k=1}^K z_0^k$  and the constant  $\rho$  denotes the “collective” risk aversion of all the market makers, defined as:

<sup>14</sup> Equation (4.20) corresponds to equation (3.44) in Chapter 3. Beside some slight difference in notation (in particular the use of  $\sigma_v^2$  instead of  $\sigma_z^2$  to denote the variance of  $v$ ), the two equations differ because here the initial inventory position and the risk-aversion coefficients can differ between dealers (as indicated by their subscript  $k$ ), while in Chapter 3 they do not.

$$\rho \equiv \left( \frac{1}{\rho^1} + \dots + \frac{1}{\rho^K} \right)^{-1}. \quad (4.24)$$

For instance, if all market makers have the same absolute risk aversion  $\bar{\rho} = \rho^1 = \dots = \rho^K$  then  $\rho = \bar{\rho}/K$ . The “collective” risk aversion  $\rho$  of the market makers declines as their number  $K$  increases, because risk is more widely spread when more dealers are drawn into market-making. So the equilibrium price is:

$$p = \underbrace{\mu - \rho \sigma_v^2 Z}_{\text{midquote } m} + \underbrace{\rho \sigma_v^2}_{\lambda} q. \quad (4.25)$$

As in Chapter 3, the midquote  $m$  is inversely related to dealers’ aggregate inventory at time  $t = 0$ . Intuitively, a dealer’s marginal valuation for the security is low if his exposure to the risk of the security is already large. Moreover, the previous equation shows that the depth of the market ( $1/\lambda$ ) is inversely related to dealers’ aggregate risk aversion,  $\rho$ , and the riskiness of the asset payoff,  $\sigma_v^2$ . So even though dealers are competitive, more dealers means a deeper market when dealers are risk averse, because the higher the number of dealers, the smaller the position that each individual dealer must take to absorb a given market order. Thus, each dealer requires less compensation for bearing inventory risk, and the market is more liquid.

As in Section 4.2.4, we can compute the effective bid-ask spread for an order of size  $|q|$ . Using equation (4.25), we obtain:

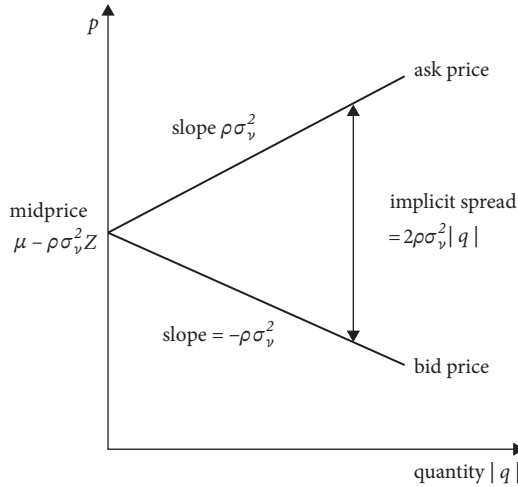
$$s = 2\rho\sigma_v^2|q|, \quad (4.26)$$

which is proportional to the size of the order imbalance as in the Kyle model (Figure 4.5).

Note that the actual price paid in the call market by any particular investor  $i$  depends not just on his own order  $q^i$  but on the total order flow from other traders,  $q - q^i$ . From the standpoint of trader  $i$ , the aggregate order flow  $q$  may be random, and the variance of the price that he will pay is  $\rho^2 \sigma_v^4 \text{var}(q|q^i)$ , where  $\text{var}(q|q^i)$  is his estimate of the variance of the total order flow from other investors. So in the call market, as the execution price is not certain ex-ante, investors face an execution risk whose size increases with trading costs: there is a positive relationship between illiquidity and execution price volatility.

### 4.3.2 Imperfectly Competitive Dealers

The previous section posits that dealers behave competitively: they ignore the effect of their own orders on the clearing price. When the number of dealers



**Figure 4.5** Competitive call market price with risk averse market makers.

is small, however, this assumption is unrealistic. Suppose for instance that a buy market order arrives. By reducing the number of shares that he offers for sale in the auction, a dealer drives the clearing price up and obtains a better execution price. Of course, the cost of this strategy is that he receives fewer shares at the equilibrium price. Hence, each dealer faces a trade-off between price and quantity, as in models of imperfect competition between firms (and very much as in the model of imperfect competition considered in Section 4.2.4). To understand this trade-off and its effect on market depth more clearly, we now relax the assumption that dealers are price takers. For simplicity, we assume that all the  $K$  market makers have the same coefficient of risk aversion  $\bar{\rho}$  and thus, combined risk aversion  $\rho = \bar{\rho}/K$ .<sup>15</sup>

To analyze the call market equilibrium in this case, we need to account for the interdependence of dealers' bidding strategies. For instance, if a dealer offers a large number of shares at a given price, other dealers will be more inclined to reduce their own orders to obtain a better execution price. We again adopt the concept of Nash equilibrium, which in this case is a set of price schedules  $\{Y^k(p)\}_{k=1}^K$  for the dealers such that the schedule of each one maximizes his expected utility given the price schedule chosen by his competitors.

In Appendix B, we show that, if  $K \geq 3$ , the following price schedules form a Nash equilibrium:

<sup>15</sup> The case where market makers are not all equally risk averse ( $\rho^1 \neq \rho^2 \neq \dots \neq \rho^K$ ) is analyzed by Röell (1998), who shows that for any given aggregate risk tolerance ( $1/\rho$ ), more equal distribution of risk tolerance among them improves market liquidity.

$$Y^k(p) = \frac{K-2}{K-1} \left( \frac{p-\mu}{\rho\sigma_v^2} + z_0^k \right). \quad (4.27)$$

As a consequence, the equilibrium price is a linear function of the market order size  $q$ :

$$p = \mu - \rho\sigma_v^2 Z + \left( \frac{K-2}{K-1} \rho\sigma_v^2 \right) q. \quad (4.28)$$

The technique used to derive this equilibrium is very similar to that employed in Appendix A. Indeed, the nature of the competition between dealers in the models with asymmetric information and inventory risk is identical. The only difference is that in the former case dealers need to infer information from the clearing price, an effect that is absent here.

Thus, in equilibrium, at each price dealers offer to sell or buy only a fraction  $\frac{K-2}{K-1}$  of the number of shares that they would be willing to trade when they behave competitively (compare equations (4.21) and (4.27)). Intuitively, by restricting the size of his order at each price, the dealer shifts the execution price to his advantage, as explained in the introduction of this section.

As a consequence, the aggregate supply schedule is less elastic, so that the price impact of a market order of a given size is larger, that is,  $\lambda$  is greater than in the competitive case. Imperfect competition among dealers reduces the depth of the market, regardless of whether the illiquidity arises from adverse selection or from inventory risk. Now, if the number of dealers increases the market becomes deeper ( $\lambda$  decreases) for *two* reasons: (i) the risk-bearing capacity of the market increases, as in the case of perfectly competitive dealers; and (ii) competition intensifies with the number of dealers, whose scope for affecting the equilibrium price by shading their offers is reduced—an effect captured by the fact that the ratio  $(K-1)/(K-2)$  is declining in  $K$ . This implies that measures of market illiquidity should be inversely related to the number of dealers in a stock, since a larger number of dealers fosters competition and increases overall risk-bearing capacity.

When the number of dealers becomes infinite, the equilibrium converges to that of the benchmark case, considered in Section 4.3.1. Thus, in presence of either asymmetric information or inventory risk, dealers' rents vanish when the number of dealers is very large (strictly speaking, infinite). This observation has important implications. First, it suggests that in many real-world situations liquidity suppliers' rents should be an important determinant of market illiquidity.<sup>16</sup> As Section 3.4.2 explains, this determinant is often bundled with

<sup>16</sup> Biais, Bisière, and Spatt (2010) show empirically that competition among liquidity suppliers is imperfect and provide estimates of their rents on one U.S. electronic market.

order-processing costs in empirical analyses. Yet for policy intervention, it is important to distinguish the two causes of illiquidity. Second, trading rules should foster competition among liquidity providers. For instance, rules that restrict the number of market makers in a security are likely to result in excessive rents and illiquidity.

Finally, in the model with imperfect competition, the expression for the midprice is the same as in the benchmark case. This means that imperfect competition does not affect the conclusions set out in Section 3.5.3 of Chapter 3 regarding the dynamics of quotes and inventories. In fact, those conclusions derive from the relationship between the mid-price and dealers' aggregate inventory, which, as noted, is unaltered by the level of dealers' market power.

#### 4.4 Further Reading

Our analysis in Section 4.2 is based on Kyle (1985). The model of imperfect competition among dealers developed in Section 4.2.4 is based on Kyle (1989), but that work considers a more general case in which bidders are both risk averse and informed. Pagano (1989a) also considers the case of imperfect competition among risk-averse dealers but in a model without asymmetric information, as in Section 4.3.2. Madhavan (1992) compares the properties of dealer markets and call markets under assumptions similar to those used in Sections 4.2.2 and 4.2.4 above. Rostek and Weretka (2015) consider a model of imperfect competition between risk averse dealers similar to that in Section 4.3.2 but use a different method to solve for the equilibrium and, more importantly, consider the case in which dealers can trade over multiple periods (see Vayanos (1999), Du and Zhu (2017), and Duffie and Zhu (2017) for related models). At a more general level, the analysis of imperfect competition among dealers is related to papers on competition in price schedules (e.g., Klemperer and Meyer 1989) and auctions of shares (Wilson 1979).

Many works have used the framework proposed by Kyle (1985) to study a variety of issues. For instance, Chowdhry and Nanda (1991) analyze multi-market trading (see Chapter 7), and Subrahmanyam (1991a) studies the effects of basket trading. Subrahmanyam (1991b) also analyzes the case of risk averse informed investors and market makers in this framework.

The analysis of Sections 4.3 and 4.2.4 emphasizes the importance of modeling imperfect competition among liquidity providers, in that dealers' rents are one cause of market illiquidity. Ho and Stoll (1983) study a model of price competition among dealers in the presence of inventory risk, observing that the best ask price should be set by the dealer with the largest position and the

best bid price by the dealer with the smallest position, since there is an inverse relationship between a dealer's bid-ask quotes and his inventory (see Chapter 3). They solve for equilibrium quotes assuming that dealers know each other's inventory (see exercise 9). Biais (1993) relaxes this assumption and, applying the revenue equivalence theorem of auction theory, finds that the average bid-ask spread should be the same whether dealers can observe the inventories of their competitors or not. Yin (2005) considers the effect of search costs in the model of Biais (1993). Biais, Foucault, and Salanié (1997) compare the outcome of competition between dealers with inventory risk in three different trading mechanisms: a floor market, a limit order market, and a dealer market. They find that the limit order market can produce the competitive equilibrium even when the number of dealers is finite, provided that market orders are allocated among dealers tied at the same price on a pro-rata basis (see also Viswanathan and Wang 2002). Biais, Martimort, and Rochet (2000) present a model of imperfect competition between liquidity suppliers in presence of asymmetric information in a discriminatory auction (a market structure that we analyze in Chapter 6).

#### 4.5 Appendix A. Adverse Selection with Imperfect Competition: Derivations

We show that if  $K > 3$ , under the assumptions of the model in Section 4.2.4, there is a rational expectations equilibrium in which:

$$Y^k(p) = \phi(p - \mu), \forall k \in \{1, \dots, K\}, \quad (4.29)$$

and the equilibrium price mapping is:

$$p^*(q) = \mu + \lambda q, \quad (4.30)$$

with  $\phi = \frac{1}{\alpha} \frac{K-2}{K(K-1)}$ ,  $\lambda = \alpha \frac{K-1}{K-2}$  and  $\alpha = \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2}$ .

We first consider the residual demand that a dealer faces for any given total market demand  $q$ , when his competitors use the equilibrium supply strategies of equation (4.29). Let  $y^k$  be the number of shares offered by dealer  $k$ . Market clearing implies that:

$$q = y^k + (K-1)\phi(p - \mu). \quad (4.31)$$

That is,

$$P(y^k) = \mu + \frac{q - y^k}{(K-1)\phi}, \quad (4.32)$$

is the “residual demand function” faced by dealer  $k$ .

We next make an educated guess about what the dealer knows. Even though he does not observe  $q$  directly, we can see that he can infer it from the market price because for any given value of his supply  $y^k$ , there is a one-to-one relationship between  $q$  and the market price given by (4.32). Accordingly, we imagine that he knows  $q$  and assume that for each given value of  $q$ , with its corresponding residual demand function and the corresponding valuation  $E(v|q) = \mu + \alpha q$ , he picks a price-quantity pair to maximize his expected profit:

$$\max_{y^k} [p - (\mu + \alpha q)],$$

where:

$$p = P(y^k) = \mu + \frac{q - y^k}{(K-1)\phi}.$$

The dealer is assumed to act strategically, and thus to be aware that the clearing price decreases by  $\frac{\partial p}{\partial y^k} = -\frac{1}{(K-1)\phi}$  if he increases his supply  $y^k$  by one unit. Hence, taking the first-order condition, we obtain:

$$\frac{q - 2y^k}{(K-1)\phi} - \alpha q = 0.$$

Thus dealer  $k$ 's price-quantity pair for any given  $q$  is:

$$\begin{aligned} y^k &= \frac{1}{2}q(1 - \alpha(K-1)\phi), \\ p &= \mu + \frac{1}{2} \left[ \frac{1}{(K-1)\phi} + \alpha \right] q. \end{aligned}$$

Eliminating  $q$  from these two equations, we find that, as the underlying  $q$  varies, dealer  $k$ 's supply function is traced out:

$$Y^k(p) = \frac{(K-1)\phi[1 - \alpha(K-1)\phi]}{1 + \alpha(K-1)\phi}(p - \mu). \quad (4.33)$$

If the equilibrium is symmetric and all market makers use supply function (4.29), the slope of the supply functions (4.29) and (4.33) must be the same:

$$\phi = \frac{(K-1)\phi[1-\alpha(K-1)\phi]}{1+\alpha(K-1)\phi},$$

which yields:

$$\phi = \frac{1}{\alpha} \frac{K-2}{K(K-1)}.$$

Then market clearing implies:

$$\sum_k \frac{1}{\alpha K} \frac{K-2}{K-1} (p^* - \mu) = q,$$

so that in equilibrium the market price function is:

$$p^* = \mu + \alpha \frac{K-1}{K-2} q.$$

Hence, as claimed,

$$\lambda = \alpha \frac{K-1}{K-2},$$

where  $\alpha = \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2}$  as defined in equation (4.4).

## 4.6 Appendix B. Inventory Risk with Imperfect Competition: Derivations

We show that if  $K > 3$ , under the assumptions of the model in Section 4.3.2, there is a Nash equilibrium in which:

$$Y^k(p) = \phi(p - \mu) + \varphi z_0^k, \forall k \in \{1, \dots, K\}, \quad (4.34)$$

with  $\phi = \frac{K-2}{K-1} \frac{1}{\bar{\rho} \sigma_z^2}$  and  $\varphi = \frac{K-2}{K-1}$ .

The reasoning is very similar to that in Appendix A. In particular, we can write the residual demand function facing market maker  $k$ : to clear the market, for any given value of total demand  $q$ , his supply  $y^k$  must equal total demand less the supply from his  $K-1$  peers, that is,

$$y^k = q - (K-1)\phi(p - \mu) - \varphi \sum_{j \neq k} z_0^j, \quad (4.35)$$

or equivalently,

$$P(y^k) = \mu + \frac{1}{(K-1)\phi} \left[ q - y^k - \varphi \sum_{j \neq k} z_0^j \right], \quad (4.36)$$

so that the sensitivity of the residual demand function faced by a dealer to his supply is again:  $\frac{\partial P}{\partial y^k} = -\frac{1}{(K-1)\phi}$ . Dealer  $k$  chooses his supply  $y^k$  to maximize his mean-variance utility:

$$\max_{y^k} \mu z^k + y^k(p - \mu) - \frac{\bar{\rho}\sigma_v^2}{2}(z_0^k - y^k)^2, \quad (4.37)$$

where  $p$  is given by (4.36). The first-order condition for this maximization problem is:

$$\frac{1}{\phi(K-1)} \left[ q - 2y^k - \varphi \sum_{j \neq k} z_0^j \right] + \bar{\rho}\sigma_v^2(z_0^k - y^k) = 0.$$

Using the market clearing condition (4.35) to substitute for  $q$ , we obtain:

$$\frac{\left[ y^k + (K-1)\phi(p - \mu) + \varphi \sum_{j \neq k} z_0^j - 2y^k - \varphi \sum_{j \neq k} z_0^j \right] + \bar{\rho}\sigma_v^2(z_0^k - y^k)}{\phi(K-1)} = 0,$$

which yields dealer  $k$ 's supply schedule:

$$Y^k(p) = \frac{1}{\frac{1}{\phi(K-1)} + \bar{\rho}\sigma_v^2} (p - \mu) + \frac{\bar{\rho}\sigma_v^2}{\frac{1}{\phi(K-1)} + \bar{\rho}\sigma_v^2} z_0^k. \quad (4.38)$$

By equating the coefficients of equation (4.38) with the initially conjectured supply schedule (4.34), it follows that there is a symmetric equilibrium of the form:

$$Y^k(p) = \frac{K-2}{K-1} \left[ \frac{1}{\bar{\rho}\sigma_v^2} (p - \mu) + z_0^k \right]. \quad (4.39)$$

To see how the market price varies with demand  $q$ , notice that market clearing requires:

$$\sum_{k=1}^K Y^k(p) = \sum_{k=1}^K \frac{K-2}{K-1} \left[ \frac{1}{\rho\sigma_v^2} (p - \mu) + z_0^k \right] = q,$$

which yields:

$$p = \mu - \rho\sigma_v^2 Z + \frac{K-1}{K-2} \rho\sigma_v^2 q,$$

as claimed in equation (4.28) in the text.

## 4.7 Exercises

1. **Kyle's model with an imperfectly informed investor.** Suppose that in the model presented in Section 4.2 (Kyle 1985), the informed investor observes a noisy signal  $s = v + \eta$  about the final value  $v$  of the security, where the noise component  $\eta \sim N(0, \sigma_\eta^2)$  has no correlation either with the security's value  $v$  or with the noise traders' order  $u$  ( $\text{cov}(\eta, v) = \text{cov}(\eta, u) = 0$ ).
  - a. Assume that competitive market makers post the following price schedule:

$$p(q) = \mu + \lambda q, \quad (4.40)$$

where  $q$  is the net order flow. Find the optimal value of  $\lambda$  that they will choose if they conjecture that the informed trader's strategy is the following function of his noisy signal  $s$ :

$$X(s) = \beta(s - \mu). \quad (4.41)$$

How does the market depth  $1/\lambda$  chosen by market makers respond to changes in the variance  $\sigma_\eta^2$  of the informed investor's error? What is the intuitive explanation for this result? If we plot the depth  $1/\lambda$  as a function of the aggressiveness  $\beta$  of informed investors (the former on the vertical axis and the latter on the horizontal axis), how do changes in the variance  $\sigma_\eta^2$  affect the position and shape of this curve?

- b. In this case, the informed trader too must solve an inference problem in forming his expectation of the security's value. Show that his expectation of  $v$  conditional on the signal  $s$  is:

$$E(v|s) = \mu + \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\eta^2} (s - \mu),$$

and find the value of  $\beta$  as a function of market depth  $1/\lambda$ . How does the trading aggressiveness  $\beta$  that informed investors choose respond to changes in the variance  $\sigma_\eta^2$  of the error term  $\eta$ ? What is the intuitive explanation for this result? If we plot  $\beta$  as informed investors' best response to the depth  $1/\lambda$  chosen by market makers in the same graph described under point **a**, how do changes in the variance  $\sigma_\eta^2$  affect the position and shape of this line?

- c. Compute the equilibrium values of  $\lambda$  and  $\beta$ . How do they respond to changes in the variance  $\sigma_\eta^2$  of the error made by the informed investor? Graphically, does the equilibrium still correspond to the point of minimum depth as in the baseline version of Kyle's model?
- d. Compute the ex ante expected profit of the informed investor. What is the effect of an increase in  $\sigma_\eta^2$  on this profit? What is the intuitive explanation for this result?

**2. Noise trading in Kyle's model.** Consider the model presented in Section 4.2 (Kyle 1985), changing only the assumption about noise trading: assume that there are two groups (1 and 2) of noise traders, whose orders are respectively  $u_1 \sim N(0, \sigma_{u_1}^2)$  and  $u_2 \sim N(0, \sigma_{u_2}^2)$ , both uncorrelated with the asset's future value (i.e.  $\text{cov}(v, u_1) = \text{cov}(v, u_2) = 0$ ). All the other assumptions of the model are unchanged: (i) market makers are risk neutral and perfectly competitive, (ii) the asset value is  $v \sim N(\mu, \sigma_v^2)$ , (iii) the informed investor's order is  $x = \beta(v - \mu)$ , and (iv) market makers only observe the total net order  $q = x + u_1 + u_2$ . [Hint: think whether your answers require computing the results of the Kyle model again.]

- a. Derive (i) the price schedule  $p = \mu + \lambda q$  chosen by competitive market makers, under the assumption that the informed trader's order is  $x = \beta(v - \mu)$ ; (ii) the optimal informed trader's order, given the price schedule chosen by market makers; and (iii) the equilibrium values of  $\lambda$  and  $\beta$ .
- b. Now suppose that market makers can observe the actual realization of the order placed by group 1 (for instance, because these are local investors, while those of group 2 are foreign investors) before trading occurs. Under this further assumption, derive again (i) the price schedule  $p = \mu + \lambda q$  chosen by competitive market makers, under the assumption that the informed trader's order is  $x = \beta(v - \mu)$ ; (ii) the optimal informed trader's order, given the price schedule chosen by market makers; and (iii) the equilibrium values of  $\lambda$  and  $\beta$ .
- c. Compare the results obtained under **a** and under **b**: does the assumption made under **b** change the equilibrium market depth and informed traders' aggressiveness and, if so, what is the intuitive reason for this difference?

- 3. Kyle's model with multiple informed traders.** Consider the model presented in Section 4.2 (Kyle 1985), but assume that there are  $N > 1$  informed traders, who all perfectly observe the final value of the security  $v$  but not the equilibrium price at the time that they determine their quantity demanded  $q_i$ .
- a. Suppose that market makers post the price schedule described by equation (4.40), where  $q$  is the net order flow  $\sum_{i=1}^N x_i + \mu$  and  $\mu = E(v)$ . Assuming that each informed trader uses the following order submission strategy:

$$x_i = X_i(v) = \beta(v - \mu) \text{ for } i \in \{1, \dots, N\},$$

find the value of  $\beta$  for which a Nash equilibrium exists, determine how  $\beta$  is affected by  $N$ , and explain intuitively why.

- b. Suppose now that investors follow the order submission strategy derived in step a. Show that in this case the market makers' pricing strategy is given by equation (4.40), and find the value of  $\lambda$  that they optimally choose.
- c. What is the market depth in equilibrium, and how is it affected by an increase in the number of informed traders,  $N$ ? What is the economic intuition for this result? Do you think that this result is robust; that is, does it still hold if the assumptions of the model are relaxed? (For instance, discuss informally whether you would still expect this result to hold if informed traders were risk averse.)
- d. Compute the ex ante expected profit of each informed investor. What is the effect of an increase in  $N$  on the aggregate profit of informed investors?
- 4. Variance of price change and average pricing error in Kyle's model.** Consider the static model by Kyle (1985), where (i) market makers are risk neutral and perfectly competitive, (ii) the asset value is  $v \sim N(\mu, \sigma_v^2)$ , (iii) the informed investor's order is  $x = \beta(v - \mu)$ , and the noise traders' order is  $u \sim N(0, \sigma_u^2)$ ; and (iv) market makers only observe the total net order  $q = x + u$ .
- a. Based on the price schedule  $p = \mu + \lambda q$  chosen by competitive market makers in this model, show that the variance of price changes is:

$$\text{var}(v - p) = \frac{\sigma_u^2}{\beta^2 \sigma_v^2 + \sigma_u^2} \sigma_v^2,$$

and explain intuitively why it is decreasing in the informed traders' aggressiveness  $\beta$ .

- b. Show that the average pricing error is:

$$\text{var}(p - \mu) = \frac{\beta^2 \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2} \sigma_v^2.$$

Is this expression increasing or decreasing in informed traders' aggressiveness  $\beta$ ? Explain intuitively why.

5. **Informed investor with price-contingent orders.** Suppose that in the Kyle (1985) model the informed investor is allowed to set a demand schedule conditioned on price rather than a market order.

- a. Show that the equilibrium is essentially the same as when the informed investor must place a market order, in the sense that average noise trader transaction costs and informed investor profits are the same. The only difference is that liquidity is provided in part by the informed investor as well as by the competitive market makers. [Hint: imagine that the informed investor submits a demand  $x$  after observing the uninformed trader demand  $u$ .]
- b. What happens under this new assumption if there is more than one informed investor?

6. **Risk-averse informed investor.** Suppose that the informed investor is risk averse (with constant coefficient of absolute risk aversion  $b$ ) and that he liquidates any amount of the security that he buys at a liquidation value  $v + \varepsilon$ , where  $\varepsilon$  is a normally distributed random variable with mean zero and variance  $\sigma_\varepsilon^2$ . At the time of trading the informed investor knows the realization of  $v$  but not that of  $\varepsilon$ . This noise in his signal implies that in taking long or short positions based on his privileged information, he bears some risk. Show that in this case his trading intensity is reduced relative to the model with risk neutrality, so that the  $\beta$  function swivels to the left (in Figure 4.3 of the text), increasing the equilibrium liquidity.

7. **Extension of the Glosten-Milgrom model to multiple trade sizes.** Consider the following simple extension of the high-low-value adverse selection model of bid-ask spreads to two trade sizes, as in Easley and O'Hara (1987). A security's true value is  $v$ , which may be high ( $v^H$ ) or low ( $v^L$ ) with probability  $\frac{1}{2}$  each. Market makers do not know the true value. In a trading period one trader comes to the market. Market makers cannot tell who the trader is:

- With probability  $1 - \pi$ , he is a "noise trader": either a "retail" customer, who wants to trade one unit, or a "wholesale" customer, who always trades two units. Half the noise traders are retail customers, and half are wholesale customers. Both types of noise trader want to buy or sell the security with probability  $\frac{1}{2}$  each.

- With probability  $\pi$ , the trader is an “insider” who knows the security’s true value  $v$ . He will buy one or two units if the true value is  $v^H$  and sell one or two units if the true value is  $v^L$ .

Ask and bid prices are set by competitive, risk-neutral market makers.

- Suppose that the insider always chooses to trade two units. What will be the ask and bid prices quoted by market makers for one unit, that is,  $a(1)$  and  $b(1)$ ? In this case, will there be a non-zero bid-ask spread for this trade size? What bid and ask prices correspond to the two-unit trade size, that is  $a(2)$  and  $b(2)$ , under the same assumption?
  - Would the insider always want to trade two units rather than one? Consider this question under two assumptions regarding  $\pi$ :  $\pi = \frac{1}{2}$  and  $\pi = \frac{1}{4}$ . In both cases, compute the insider’s profit if he trades two units and compare it with his profit when he trades one unit.
  - Do you have any ideas on what would happen in situations where your answer to point **b** is “No”? In particular, could it be an equilibrium for the insider to always trade one unit? [Hard question: characterize the equilibrium in which the insider always chooses his trade size to maximize his profit!]
- 8. Informed traders’ optimal trade size with imperfectly competitive dealers.** Consider the model of imperfect competition among dealers in Section 4.2.4. Show that informed investors’ trade size in this model is:  $x(v) = \beta(v - \mu)$  with  $\beta = \sqrt{\frac{K-2}{K}} \cdot \frac{\sigma_u}{\sigma_v}$ . Why does the informed investor’s trade size increase with the number of dealers,  $K$ ?
- 9. Competition between dealers with different initial inventories.** This exercise analyzes (along the lines of the model by Ho and Stoll 1983) how quotes are determined in a competitive dealer market where dealers have different inventories. Consider the model with inventory risk considered in Section 3.5.1. Instead of a representative dealer, assume that there are two dealers, 1 and 2, with respective initial inventories  $z_1$  and  $z_2$ , where  $z_1 \geq z_2$ . These dealers have mean-variance preferences with the same risk aversion,  $\rho$ . An investor contacts the dealers to execute a buy or a sell order of size  $q$ . The investor requests a quote simultaneously from both dealers and directs his order to that posting the best offer. If dealers make the same offer, the investor routes his order to either one with probability  $\frac{1}{2}$ .
- Suppose that the investor contacts the dealers to execute a buy order. Compute the reservation ask prices, denoted  $a^r(z_1)$  and  $a^r(z_2)$ , that make dealer 1 and 2 indifferent between selling  $q$  shares or not. Explain why  $a^r(z_1) \leq a^r(z_2)$ .

- b. Suppose that the investor contacts the dealers to execute a sell order. Compute the reservation bid prices,  $b^r(z_1)$  and  $b^r(z_2)$ , that make dealers 1 and 2 indifferent between buying  $q$  shares or not. Explain why  $b^r(z_2) \geq b^r(z_1)$ .
- c. What are the quotes (denoted by  $a_1^*$ ,  $a_2^*$ ,  $b_1^*$ ,  $b_2^*$ ) that will be posted by the dealers in a Nash equilibrium where each dealer knows the inventory of his competitor? [Hint: in such an equilibrium, no dealer can be better off by undercutting the quote of his competitor.]
- d. Are dealers' rents increasing or decreasing in the difference between their initial inventories?
- e. How would your answer to point c change if there is a third dealer with inventory  $z_3$  such that  $z_1 > z_2 > z_3$ ?
- 10. Risk-averse market makers active in two correlated markets.** Consider a two-period model with  $K$  competitive and identically risk-averse market makers, who in the first period operate in the markets for two assets, 1 and 2. The objective function of market maker  $k = (1, 2, \dots, K)$  is:

$$E(w^k) - \frac{\bar{\rho}}{2} \text{var}(w^k),$$

where  $\bar{\rho}$  the individual risk aversion of market makers and  $w^k$  denotes the final wealth of market maker  $k$ , which is:

$$w^k = v_1 \underbrace{(z_1^k - y_1^k)}_{\substack{\text{final} \\ \text{inventory} \\ \text{of asset 1}}} + v_2 \underbrace{(z_2^k - y_2^k)}_{\substack{\text{final} \\ \text{inventory} \\ \text{of asset 2}}} + \underbrace{p_1 y_1^k + p_2 y_2^k + c^k}_{\substack{\text{final} \\ \text{cash} \\ \text{position}}},$$

where  $p_j$  denotes the price of asset  $j$ ,  $z_j^k$  the initial inventory of asset  $j$  held by market maker  $k$ , and  $y_j^k$  is the supply of asset  $j$  by market maker  $k$ . The final values of assets 1 and 2,  $v_1$  and  $v_2$ , have identical mean and variance, and correlation  $r$ :

$$E(v_1) = E(v_2) = \mu, \text{var}(v_1) = \text{var}(v_2) = \sigma_v^2, \text{cov}(v_1, v_2) = r\sigma_v^2.$$

- a. Derive the optimal supply of the two assets by market maker  $k$ .
- b. Denoting the total inventories and net demand for asset 1 by  $Z_1$  and  $q_1$  respectively, and those for asset 2 by  $Z_2$  and  $q_2$ , derive the equilibrium price of the two assets. (Recall that, since market makers are all

identically risk averse, the “collective” risk aversion in the market can be written as  $\rho = \bar{\rho}/K$ .)

- c. How does the equilibrium price of each asset respond to the net order flow of the other? How does that depend on the correlation between the two asset values,  $r$ ? Would this hold if two separate sets of market makers were active in each of the two markets? Why, intuitively?

**11. Equilibrium number of risk-averse market-makers in a competitive call auction market.** Consider the two-period call auction model with  $K$  competitive and identically risk-averse market makers. In that setting, each market maker  $k = (1, 2, \dots, K)$  maximizes the mean-variance objective:

$$U^k = E(w^k) - \frac{\bar{\rho}}{2} \text{var}(w^k),$$

where  $\bar{\rho}$  is the risk aversion coefficient and  $w^k$  is the final wealth of market maker  $k$ . For simplicity, all market makers are assumed to have an identical initial inventory ( $z^k = z$ ) of the risky asset and a zero initial cash position ( $c^k = 0$ ), so that their final wealth is:

$$w^k = v(z - y^k) + p y^k,$$

where  $p$  denotes the price of the risky asset and  $y^k$  the supply of the asset by market maker  $k$ , so that  $z - y^k$  is his final inventory.

Now suppose that, in order to participate to trading, each market maker must pay a fixed cost  $C > 0$ , which can be taken to reflect the resource cost of engaging in trading (computer equipment, employees' pay, etc.) and/or that arising from financial frictions (e.g., borrowing costs). Hence, we must consider a *participation constraint*: market maker  $k$  participates to the market only if in equilibrium the resulting value of his objective function is at least as large as a threshold level of utility  $\bar{U}$ , which denotes his utility if he does not supply the asset ( $y^k = 0$ ), thus retaining his whole initial inventory  $z$ , and saves the cost  $C$ :

$$U^k \geq \bar{U} = E(v \cdot z) - \frac{\bar{\rho}}{2} \text{var}(v \cdot z) + C = \left( \mu - \frac{\bar{\rho}}{2} \sigma_v^2 z \right) \cdot z + C.$$

Denote by  $K^*$  the number of market makers who *in equilibrium participate* to the market by supplying a non-zero amount  $y^k$  of the risky asset. To help answering the subsequent questions, recall that, if market maker  $k$  participates to the market, his optimal supply of the risky asset is:

$$y^k = \frac{p - \mu}{\bar{\rho}\sigma_v^2} + z,$$

where  $\mu$  and  $\sigma_v^2$  are the mean and the variance of the asset's value, and that the equilibrium price results from equating total supply with net demand  $q$ :

$$\sum_{k=1}^{K^*} y^k = q \Rightarrow p^* = \mu - \rho\sigma_v^2 Z + \rho\sigma_v^2 q = \mu - \bar{\rho}\sigma_v^2 z + \bar{\rho}\sigma_v^2 \frac{q}{K^*},$$

where  $Z$  denotes the total inventory of the  $K^*$  market makers who participate to the market and  $\rho$  their “collective” risk aversion:  $Z = K^* z$  and  $\rho = \bar{\rho}/K^*$ .

- a. Find the equilibrium value of the objective function  $U^k$  for a market maker  $k \in S^*$  who participates to the market.
- b. Determine the marginal market maker, i.e. the market maker for which the participation constraint  $U^k \geq \bar{U}$  just binds (in the sense that the constraint would be violated by adding another market maker to the set of participants), and the number  $K^*$  of market makers for which the participation constraint  $U^k \geq \bar{U}$  is satisfied in equilibrium (neglecting integer constraints, for simplicity).
- c. How does  $K^*$  vary with  $\sigma_v^2$ ,  $z$  and  $q$ ? Why, intuitively?

**12. Inventory effects around treasury auctions.** The data for this exercise are contained in the Excel file Ch4\_ex12\_data.xlsx, available on the companion website for the book.<sup>17</sup> The file provides data on 105 auctions of French long-term government bonds from 2010 to 2021. The auctions are run by the national debt management office (Agence France Trésor) and the participants are mainly banks, who are active as dealers in the secondary markets for government bonds.

Each row in the Excel file contains the notional amount of bonds issued and the benchmark 30-year yield 5 days before the auction, on the day of the auction, and 5 days after the auction (i.e.,  $i_{t-5}$ ,  $i_t$ , and  $i_{t+5}$ , where  $i_t$  is the bond yield at the auction date). The bond yield is equal to the return from holding the bond to maturity. Since bonds' cash flows are fixed, the yield is inversely related to the bond price: the yield falls when the price of the bond rises,

<sup>17</sup> Sources: [www.aft.gouv.fr](http://www.aft.gouv.fr) and Bloomberg.

- a. As dealers buy bonds at auction, the auction will cause their inventory to increase. Based on the inventory risk model (see equation (4.25)), what effect should we then expect the auction to have on the benchmark yield (recalling that yields and prices move inversely)? Compute the average yield change relative to 5 days before the auction. Is the evidence in line with the model's prediction?
- b. Dealers usually decrease their bond holdings in the days following the auction by selling bonds from their inventory to their customers. Which effect should this have on the yield? Is this visible in the data?
- c. Based on your replies to questions **a** and **b**, could you devise a trading strategy to exploit the observed pattern in prices? These patterns have actually been observed in many different government bond markets (see Lou, Yan, and Zhang 2013). Auctions are also announced in advance, so the auction itself does not bring any new information. Using the dynamic model presented in Section 3.5.2 (Chapter 3), can you explain why such predictable patterns can persist?